In [1]:
```
import pandas as pd
import numpy as np

df = pd.read_csv("ecommerce_dirty_dataset_15_records.csv")

df.head()
```

Out[1]: Columns 1–8 of 10

	Order_ID	Order_Date	Product_Title	Category	Quantity	Unit_Price	Customer_Email
0	1	03-01-2026	Wireless Mouse	Electronics	2	599	asha@example.com
1	2	04-01-2026	USB-C Cable	electronics	3	199	ravi@example.com
2	3	05-01-2026	Bluetooth Speaker	Audio	1	2499	meena@example.com
3	4	06-01-2026	Laptop Stand	Accessories	NaN	899	kiran@example.com
4	5	07-01-2026	Wireless Keyboard	Electronics	two	1299	divya@example.com

Columns 9–10 of 10 · first column repeated

	Payment_Status	Rating
0	Paid	5.0
1	paid	4.0
2	Completed	5.0
3	Paid	4.0
4	Pending	3.0

In [2]:
```
df.isnull().sum()
```

Out[2]:
```
Order_ID          0
Order_Date        0
Product_Title     1
Category          0
Quantity          1
Unit_Price        1
Customer_Email    1
Payment_Status    0
Rating            1
dtype: int64
```

In [3]:
```
df.duplicated().sum()
```

Out[3]:
```
np.int64(1)
```

In [4]:
```
df[df.duplicated()]
```

Out[4]: Columns 1–8 of 10

	Order_ID	Order_Date	Product_Title	Category	Quantity	Unit_Price	Customer_Email
15	2	04-01-2026	USB-C Cable	electronics	3	199	ravi@example.com

Columns 9–10 of 10 · first column repeated

	Payment_Status	Rating
15	paid	4.0

In [5]:
```
df = df.drop_duplicates()
```

In [6]:
```
df["Product_Title"] = df["Product_Title"].str.strip()
```

In [7]:
```
df["Quantity"] = pd.to_numeric(
    df["Quantity"], errors="coerce"
)
```

In [9]:
```
df["Quantity"].info()
```

```
<class 'pandas.Series'>
RangeIndex: 15 entries, 0 to 14
Series name: Quantity
Non-Null Count  Dtype
--------------  -----
13 non-null     float64
dtypes: float64(1)
memory usage: 252.0 bytes
```

In [10]:
```
df.info()
```

```
<class 'pandas.DataFrame'>
RangeIndex: 15 entries, 0 to 14
Data columns (total 9 columns):
 #   Column          Non-Null Count  Dtype
---  ------          --------------  -----
 0   Order_ID        15 non-null     int64
 1   Order_Date      15 non-null     str
 2   Product_Title   14 non-null     str
 3   Category        15 non-null     str
 4   Quantity        13 non-null     float64
 5   Unit_Price      14 non-null     str
 6   Customer_Email  14 non-null     str
 7   Payment_Status  15 non-null     str
 8   Rating          14 non-null     float64
dtypes: float64(2), int64(1), str(6)
memory usage: 1.2 KB
```

In [11]:
```
df["Unit_Price"] = pd.to_numeric(
    df["Unit_Price"], errors="coerce"
)
```

In [13]:
```
df["Unit_Price"].info()
```

```
<class 'pandas.Series'>
RangeIndex: 15 entries, 0 to 14
Series name: Unit_Price
Non-Null Count  Dtype
--------------  -----
13 non-null     float64
dtypes: float64(1)
memory usage: 252.0 bytes
```

In [14]:
```
df["Order_Date"] = pd.to_datetime(
    df["Order_Date"], errors="coerce", dayfirst=True
)
```

In [15]:
```
df["Order_Date"].info()
```

```
<class 'pandas.Series'>
RangeIndex: 15 entries, 0 to 14
Series name: Order_Date
Non-Null Count  Dtype
--------------  -----
14 non-null     datetime64[us]
dtypes: datetime64[us](1)
memory usage: 252.0 bytes
```

In [16]:
```
df.info()
```

```
<class 'pandas.DataFrame'>
RangeIndex: 15 entries, 0 to 14
Data columns (total 9 columns):
 #   Column          Non-Null Count  Dtype
---  ------          --------------  -----
 0   Order_ID        15 non-null     int64
 1   Order_Date      14 non-null     datetime64[us]
 2   Product_Title   14 non-null     str
 3   Category        15 non-null     str
 4   Quantity        13 non-null     float64
 5   Unit_Price      13 non-null     float64
 6   Customer_Email  14 non-null     str
 7   Payment_Status  15 non-null     str
 8   Rating          14 non-null     float64
dtypes: datetime64[us](1), float64(3), int64(1), str(4)
memory usage: 1.2 KB
```

In [17]:
```
df.shape
```

Out[17]:
```
(15, 9)
```

In [18]:
```
df.dtypes
```

Out[18]:
```
Order_ID                    int64
Order_Date         datetime64[us]
Product_Title                 str
Category                      str
Quantity                  float64
Unit_Price                float64
Customer_Email                str
Payment_Status                str
Rating                    float64
dtype: object
```

In [19]:
```
df.columns
```

Out[19]:
```
Index(['Order_ID', 'Order_Date', 'Product_Title', 'Category', 'Quantity',
       'Unit_Price', 'Customer_Email', 'Payment_Status', 'Rating'],
      dtype='str')
```

In [20]:
```
df["Order_Date"].dtype
```

Out[20]:
```
dtype('<M8[us]')
```

In [21]:
```
Q1 = df["Unit_Price"].quantile(0.25)
Q3 = df["Unit_Price"].quantile(0.75)

IQR = Q3 - Q1

lower_limit = Q1 - 1.5 * IQR
upper_limit = Q3 + 1.5 * IQR

print("Q1:", Q1)
print("Q3:", Q3)
print("IQR:", IQR)
print("Lower limit:", lower_limit)
print("Upper limit:", upper_limit)
```

```
Q1: 599.0
Q3: 1599.0
IQR: 1000.0
Lower limit: -901.0
Upper limit: 3099.0
```

In [22]:
```
def classify_price(price):
    if pd.isna(price):
        return "Missing"
    elif price < lower_limit:
        return "Low Price"
    elif price > upper_limit:
        return "High Price"
    else:
        return "Medium"
```

In [23]:
```
df["Price_Category"] = df["Unit_Price"].apply(classify_price)

df[["Product_Title", "Unit_Price", "Price_Category"]]
```

Out[23]:

	Product_Title	Unit_Price	Price_Category
0	Wireless Mouse	599.0	Medium
1	USB-C Cable	199.0	Medium
2	Bluetooth Speaker	2499.0	Medium
3	Laptop Stand	899.0	Medium
4	Wireless Keyboard	1299.0	Medium
5	Phone Case	NaN	Missing
6	Power Bank	99999.0	High Price
7	HDMI Adapter	450.0	Medium
8	NaN	1299.0	Medium
9	Desk Lamp	799.0	Medium
10	Laptop Sleeve	NaN	Missing
11	Smart Watch	3499.0	High Price
12	Gaming Mouse	1599.0	Medium
13	Wireless Mouse	599.0	Medium
14	USB-C Cable	199.0	Medium

In [24]:
```
df
```

Out[24]: Columns 1–8 of 11

	Order_ID	Order_Date	Product_Title	Category	Quantity	Unit_Price	Customer_Email
0	1	2026-01-03	Wireless Mouse	Electronics	2.0	599.0	asha@example.com
1	2	2026-01-04	USB-C Cable	electronics	3.0	199.0	ravi@example.com
2	3	2026-01-05	Bluetooth Speaker	Audio	1.0	2499.0	meena@example.com
3	4	2026-01-06	Laptop Stand	Accessories	NaN	899.0	kiran@example.com
4	5	2026-01-07	Wireless Keyboard	Electronics	NaN	1299.0	divya@example.com
5	6	2026-01-08	Phone Case	Mobile	2.0	NaN	arun@example.com
6	7	2026-01-09	Power Bank	mobile accessories	1.0	99999.0	sita@example.com
7	8	NaT	HDMI Adapter	Accessories	4.0	450.0	naveen@example.com
8	9	2026-01-11	NaN	Home	1.0	1299.0	priya@example.com
9	10	2026-01-12	Desk Lamp	Home	-2.0	799.0	rahul@example.com
10	11	2026-01-13	Laptop Sleeve	Accessories	1.0	NaN	not-an-email
11	12	2026-01-14	Smart Watch	Electronics	1.0	3499.0	NaN
12	13	2026-01-15	Gaming Mouse	electronics	2.0	1599.0	john@example.com
13	14	2026-01-16	Wireless Mouse	Electronics	2.0	599.0	asha@example.com
14	15	2026-01-04	USB-C Cable	electronics	3.0	199.0	ravi@example.com

Columns 9–11 of 11 · first column repeated

	Payment_Status	Rating	Price_Category
0	Paid	5.0	Medium
1	paid	4.0	Medium
2	Completed	5.0	Medium
3	Paid	4.0	Medium
4	Pending	3.0	Medium
5	Paid	4.0	Missing
6	Paid	5.0	High Price
7	Paid	4.0	Medium
8	Cancelled	2.0	Medium

	Payment_Status	Rating	Price_Category
9	Paid	6.0	Medium
10	Piad	4.0	Missing
11	Pending	NaN	High Price
12	Paid	4.0	Medium
13	Paid	5.0	Medium
14	paid	4.0	Medium

In [25]:
```
email_pattern = r"^[^@\s]+@[^@\s]+\.[^@\s]+$"

df["Email_Valid"] = df["Customer_Email"].str.match(
    email_pattern, na=False
)
```

In [26]:
```
df[["Customer_Email"]]
```

Out[26]:

	Customer_Email
0	asha@example.com
1	ravi@example.com
2	meena@example.com
3	kiran@example.com
4	divya@example.com
5	arun@example.com
6	sita@example.com
7	naveen@example.com
8	priya@example.com
9	rahul@example.com
10	not-an-email
11	NaN
12	john@example.com
13	asha@example.com
14	ravi@example.com

In [27]:
```
df["Order_Date"] = pd.to_datetime(
    df["Order_Date"], errors="coerce", dayfirst=True
)

df.loc[df["Quantity"] ⩽ 0, "Quantity"] = np.nan
df.loc[df["Unit_Price"] < 0, "Unit_Price"] = np.nan
df.loc[~df["Rating"].between(1, 5), "Rating"] = np.nan
```

In [28]:
```
df
```

Out[28]: Columns 1–8 of 12

	Order_ID	Order_Date	Product_Title	Category	Quantity	Unit_Price	Customer_Email
0	1	2026-01-03	Wireless Mouse	Electronics	2.0	599.0	asha@example.com
1	2	2026-01-04	USB-C Cable	electronics	3.0	199.0	ravi@example.com
2	3	2026-01-05	Bluetooth Speaker	Audio	1.0	2499.0	meena@example.com
3	4	2026-01-06	Laptop Stand	Accessories	NaN	899.0	kiran@example.com
4	5	2026-01-07	Wireless Keyboard	Electronics	NaN	1299.0	divya@example.com
5	6	2026-01-08	Phone Case	Mobile	2.0	NaN	arun@example.com
6	7	2026-01-09	Power Bank	mobile accessories	1.0	99999.0	sita@example.com
7	8	NaT	HDMI Adapter	Accessories	4.0	450.0	naveen@example.com
8	9	2026-01-11	NaN	Home	1.0	1299.0	priya@example.com
9	10	2026-01-12	Desk Lamp	Home	NaN	799.0	rahul@example.com
10	11	2026-01-13	Laptop Sleeve	Accessories	1.0	NaN	not-an-email
11	12	2026-01-14	Smart Watch	Electronics	1.0	3499.0	NaN
12	13	2026-01-15	Gaming Mouse	electronics	2.0	1599.0	john@example.com
13	14	2026-01-16	Wireless Mouse	Electronics	2.0	599.0	asha@example.com
14	15	2026-01-04	USB-C Cable	electronics	3.0	199.0	ravi@example.com

Columns 9–12 of 12 · first column repeated

	Payment_Status	Rating	Price_Category	Email_Valid
0	Paid	5.0	Medium	True
1	paid	4.0	Medium	True
2	Completed	5.0	Medium	True
3	Paid	4.0	Medium	True
4	Pending	3.0	Medium	True
5	Paid	4.0	Missing	True
6	Paid	5.0	High Price	True
7	Paid	4.0	Medium	True
8	Cancelled	2.0	Medium	True

	Payment_Status	Rating	Price_Category	Email_Valid
9	Paid	NaN	Medium	True
10	Piad	4.0	Missing	False
11	Pending	NaN	High Price	False
12	Paid	4.0	Medium	True
13	Paid	5.0	Medium	True
14	paid	4.0	Medium	True

In [29]:
```
df.head(3)
```

Out[29]: Columns 1–8 of 12

	Order_ID	Order_Date	Product_Title	Category	Quantity	Unit_Price	Customer_Email
0	1	2026-01-03	Wireless Mouse	Electronics	2.0	599.0	asha@example.com
1	2	2026-01-04	USB-C Cable	electronics	3.0	199.0	ravi@example.com
2	3	2026-01-05	Bluetooth Speaker	Audio	1.0	2499.0	meena@example.com

Columns 9–12 of 12 · first column repeated

	Payment_Status	Rating	Price_Category	Email_Valid
0	Paid	5.0	Medium	True
1	paid	4.0	Medium	True
2	Completed	5.0	Medium	True

In [30]:
```
df["Category"]=df["Category"].str.strip().str.title()
```

In [31]:
```
df[["Category"]]
```

Out[31]:

	Category
0	Electronics
1	Electronics
2	Audio
3	Accessories
4	Electronics
5	Mobile
6	Mobile Accessories
7	Accessories
8	Home
9	Home
10	Accessories
11	Electronics
12	Electronics
13	Electronics
14	Electronics

In [32]:
```
df["Payment_Status"]=df["Payment_Status"].str.strip().str.title()
```

In [33]:
```
df[["Payment_Status"]]
```

Out[33]:

	Payment_Status
0	Paid
1	Paid
2	Completed
3	Paid
4	Pending
5	Paid
6	Paid
7	Paid
8	Cancelled
9	Paid
10	Piad
11	Pending
12	Paid
13	Paid
14	Paid

In [34]:
```
df
```

Out[34]: Columns 1–8 of 12

	Order_ID	Order_Date	Product_Title	Category	Quantity	Unit_Price	Customer_Email
0	1	2026-01-03	Wireless Mouse	Electronics	2.0	599.0	asha@example.com
1	2	2026-01-04	USB-C Cable	Electronics	3.0	199.0	ravi@example.com
2	3	2026-01-05	Bluetooth Speaker	Audio	1.0	2499.0	meena@example.com
3	4	2026-01-06	Laptop Stand	Accessories	NaN	899.0	kiran@example.com
4	5	2026-01-07	Wireless Keyboard	Electronics	NaN	1299.0	divya@example.com
5	6	2026-01-08	Phone Case	Mobile	2.0	NaN	arun@example.com
6	7	2026-01-09	Power Bank	Mobile Accessories	1.0	99999.0	sita@example.com
7	8	NaT	HDMI Adapter	Accessories	4.0	450.0	naveen@example.com
8	9	2026-01-11	NaN	Home	1.0	1299.0	priya@example.com
9	10	2026-01-12	Desk Lamp	Home	NaN	799.0	rahul@example.com
10	11	2026-01-13	Laptop Sleeve	Accessories	1.0	NaN	not-an-email
11	12	2026-01-14	Smart Watch	Electronics	1.0	3499.0	NaN
12	13	2026-01-15	Gaming Mouse	Electronics	2.0	1599.0	john@example.com
13	14	2026-01-16	Wireless Mouse	Electronics	2.0	599.0	asha@example.com
14	15	2026-01-04	USB-C Cable	Electronics	3.0	199.0	ravi@example.com

Columns 9–12 of 12 · first column repeated

	Payment_Status	Rating	Price_Category	Email_Valid
0	Paid	5.0	Medium	True
1	Paid	4.0	Medium	True
2	Completed	5.0	Medium	True
3	Paid	4.0	Medium	True
4	Pending	3.0	Medium	True
5	Paid	4.0	Missing	True
6	Paid	5.0	High Price	True
7	Paid	4.0	Medium	True

	Payment_Status	Rating	Price_Category	Email_Valid
8	Cancelled	2.0	Medium	True
9	Paid	NaN	Medium	True
10	Piad	4.0	Missing	False
11	Pending	NaN	High Price	False
12	Paid	4.0	Medium	True
13	Paid	5.0	Medium	True
14	Paid	4.0	Medium	True

In [37]:
```
df.info()
```

```
<class 'pandas.DataFrame'>
RangeIndex: 15 entries, 0 to 14
Data columns (total 11 columns):
 #   Column          Non-Null Count  Dtype
---  ------          --------------  -----
 0   Order_ID        15 non-null     int64
 1   Order_Date      14 non-null     datetime64[us]
 2   Product_Title   14 non-null     str
 3   Category        15 non-null     str
 4   Quantity        12 non-null     float64
 5   Unit_Price      13 non-null     float64
 6   Customer_Email  14 non-null     str
 7   Payment_Status  15 non-null     str
 8   Rating          13 non-null     float64
 9   Price_Category  15 non-null     str
 10  Email_Valid     15 non-null     bool
dtypes: bool(1), datetime64[us](1), float64(3), int64(1), str(5)
memory usage: 1.3 KB
```

In [38]:
```
text_cols = ["Product_Title", "Category", "Customer_Email",
"Payment_Status"]

for col in text_cols:
    df[col] = df[col].astype("string").str.strip()
    df[col] = df[col].replace("", np.nan)
```

In [39]:
```
df["Quantity"] = df["Quantity"].fillna(df["Quantity"].median())
df["Unit_Price"] = df["Unit_Price"].fillna(df["Unit_Price"].median())
df["Rating"] = df["Rating"].fillna(df["Rating"].median())
```

In [40]:
```
df
```

Out[40]: Columns 1–8 of 12

	Order_ID	Order_Date	Product_Title	Category	Quantity	Unit_Price	Customer_Email
0	1	2026-01-03	Wireless Mouse	Electronics	2.0	599.0	asha@example.com
1	2	2026-01-04	USB-C Cable	Electronics	3.0	199.0	ravi@example.com
2	3	2026-01-05	Bluetooth Speaker	Audio	1.0	2499.0	meena@example.com
3	4	2026-01-06	Laptop Stand	Accessories	2.0	899.0	kiran@example.com
4	5	2026-01-07	Wireless Keyboard	Electronics	2.0	1299.0	divya@example.com
5	6	2026-01-08	Phone Case	Mobile	2.0	899.0	arun@example.com
6	7	2026-01-09	Power Bank	Mobile Accessories	1.0	99999.0	sita@example.com
7	8	NaT	HDMI Adapter	Accessories	4.0	450.0	naveen@example.com
8	9	2026-01-11	<NA>	Home	1.0	1299.0	priya@example.com
9	10	2026-01-12	Desk Lamp	Home	2.0	799.0	rahul@example.com
10	11	2026-01-13	Laptop Sleeve	Accessories	1.0	899.0	not-an-email
11	12	2026-01-14	Smart Watch	Electronics	1.0	3499.0	<NA>
12	13	2026-01-15	Gaming Mouse	Electronics	2.0	1599.0	john@example.com
13	14	2026-01-16	Wireless Mouse	Electronics	2.0	599.0	asha@example.com
14	15	2026-01-04	USB-C Cable	Electronics	3.0	199.0	ravi@example.com

Columns 9–12 of 12 · first column repeated

	Payment_Status	Rating	Price_Category	Email_Valid
0	Paid	5.0	Medium	True
1	Paid	4.0	Medium	True
2	Completed	5.0	Medium	True
3	Paid	4.0	Medium	True
4	Pending	3.0	Medium	True
5	Paid	4.0	Missing	True
6	Paid	5.0	High Price	True
7	Paid	4.0	Medium	True

	Payment_Status	Rating	Price_Category	Email_Valid
8	Cancelled	2.0	Medium	True
9	Paid	4.0	Medium	True
10	Piad	4.0	Missing	False
11	Pending	4.0	High Price	False
12	Paid	4.0	Medium	True
13	Paid	5.0	Medium	True
14	Paid	4.0	Medium	True

In [41]:
```
df["Product_Title"] = df["Product_Title"].fillna("Unknown Product")
df["Category"] = df["Category"].fillna("Unknown")
df["Customer_Email"] = df["Customer_Email"].fillna("Not Provided")
df["Payment_Status"] = df["Payment_Status"].fillna("Unknown")
```

In [42]:
```
df["Order_Date"] = df["Order_Date"].fillna(df["Order_Date"].median())
```

In [43]:
```
df
```

Out[43]: Columns 1–8 of 12

	Order_ID	Order_Date	Product_Title	Category	Quantity	Unit_Price	Customer_Email
0	1	2026-01-03 00:00:00	Wireless Mouse	Electronics	2.0	599.0	asha@example.com
1	2	2026-01-04 00:00:00	USB-C Cable	Electronics	3.0	199.0	ravi@example.com
2	3	2026-01-05 00:00:00	Bluetooth Speaker	Audio	1.0	2499.0	meena@example.com
3	4	2026-01-06 00:00:00	Laptop Stand	Accessories	2.0	899.0	kiran@example.com
4	5	2026-01-07 00:00:00	Wireless Keyboard	Electronics	2.0	1299.0	divya@example.com
5	6	2026-01-08 00:00:00	Phone Case	Mobile	2.0	899.0	arun@example.com
6	7	2026-01-09 00:00:00	Power Bank	Mobile Accessories	1.0	99999.0	sita@example.com
7	8	2026-01-08 12:00:00	HDMI Adapter	Accessories	4.0	450.0	naveen@example.com
8	9	2026-01-11 00:00:00	Unknown Product	Home	1.0	1299.0	priya@example.com
9	10	2026-01-12 00:00:00	Desk Lamp	Home	2.0	799.0	rahul@example.com
10	11	2026-01-13 00:00:00	Laptop Sleeve	Accessories	1.0	899.0	not-an-email
11	12	2026-01-14 00:00:00	Smart Watch	Electronics	1.0	3499.0	Not Provided
12	13	2026-01-15 00:00:00	Gaming Mouse	Electronics	2.0	1599.0	john@example.com
13	14	2026-01-16 00:00:00	Wireless Mouse	Electronics	2.0	599.0	asha@example.com
14	15	2026-01-04 00:00:00	USB-C Cable	Electronics	3.0	199.0	ravi@example.com

Columns 9–12 of 12 · first column repeated

	Payment_Status	Rating	Price_Category	Email_Valid
0	Paid	5.0	Medium	True
1	Paid	4.0	Medium	True
2	Completed	5.0	Medium	True
3	Paid	4.0	Medium	True
4	Pending	3.0	Medium	True
5	Paid	4.0	Missing	True
6	Paid	5.0	High Price	True
7	Paid	4.0	Medium	True

	Payment_Status	Rating	Price_Category	Email_Valid
8	Cancelled	2.0	Medium	True
9	Paid	4.0	Medium	True
10	Piad	4.0	Missing	False
11	Pending	4.0	High Price	False
12	Paid	4.0	Medium	True
13	Paid	5.0	Medium	True
14	Paid	4.0	Medium	True

In [44]:
```
df["Email_Valid"] = df["Customer_Email"].str.match(
    r"^[\w.+-]+@[\w.-]+\.[A-Za-z]{2,}$",
    na=False
)
```

In [45]:
```
df
```

Out[45]: Columns 1–8 of 12

	Order_ID	Order_Date	Product_Title	Category	Quantity	Unit_Price	Customer_Email
0	1	2026-01-03 00:00:00	Wireless Mouse	Electronics	2.0	599.0	asha@example.com
1	2	2026-01-04 00:00:00	USB-C Cable	Electronics	3.0	199.0	ravi@example.com
2	3	2026-01-05 00:00:00	Bluetooth Speaker	Audio	1.0	2499.0	meena@example.com
3	4	2026-01-06 00:00:00	Laptop Stand	Accessories	2.0	899.0	kiran@example.com
4	5	2026-01-07 00:00:00	Wireless Keyboard	Electronics	2.0	1299.0	divya@example.com
5	6	2026-01-08 00:00:00	Phone Case	Mobile	2.0	899.0	arun@example.com
6	7	2026-01-09 00:00:00	Power Bank	Mobile Accessories	1.0	99999.0	sita@example.com
7	8	2026-01-08 12:00:00	HDMI Adapter	Accessories	4.0	450.0	naveen@example.com
8	9	2026-01-11 00:00:00	Unknown Product	Home	1.0	1299.0	priya@example.com
9	10	2026-01-12 00:00:00	Desk Lamp	Home	2.0	799.0	rahul@example.com
10	11	2026-01-13 00:00:00	Laptop Sleeve	Accessories	1.0	899.0	not-an-email
11	12	2026-01-14 00:00:00	Smart Watch	Electronics	1.0	3499.0	Not Provided
12	13	2026-01-15 00:00:00	Gaming Mouse	Electronics	2.0	1599.0	john@example.com
13	14	2026-01-16 00:00:00	Wireless Mouse	Electronics	2.0	599.0	asha@example.com
14	15	2026-01-04 00:00:00	USB-C Cable	Electronics	3.0	199.0	ravi@example.com

Columns 9–12 of 12 · first column repeated

	Payment_Status	Rating	Price_Category	Email_Valid
0	Paid	5.0	Medium	True
1	Paid	4.0	Medium	True
2	Completed	5.0	Medium	True
3	Paid	4.0	Medium	True
4	Pending	3.0	Medium	True
5	Paid	4.0	Missing	True
6	Paid	5.0	High Price	True
7	Paid	4.0	Medium	True

	Payment_Status	Rating	Price_Category	Email_Valid
8	Cancelled	2.0	Medium	True
9	Paid	4.0	Medium	True
10	Piad	4.0	Missing	False
11	Pending	4.0	High Price	False
12	Paid	4.0	Medium	True
13	Paid	5.0	Medium	True
14	Paid	4.0	Medium	True